UNIT 9

ATOMIC AND NUCLEAR PHYSICS

All of physics is either impossible or trivial. It is impossible until you understand it, and then it becomes trivial

– Ernest Rutherford

Learning Objectives

In this unit, the students are exposed to

- electric discharge through the gases
- determination of specific charge of by an electron J.J. Thomson experiment
- determination of electronic charge by Millikan's oil drop experiment
- atom models – J.J. Thomson and Rutherford
- Bohr atom model and hydrogen atom
- atomic spectrum and hydrogen spectrum
- structure and properties of nucleus
- various classification of nuclei based on atomic number and mass number
- mass defect and binding energy
- relation between stability and binding energy curve
- alpha decay, beta decay and gamma emission
- law of radioactive decay
- nuclear fission and fusion
- elementary ideas of nuclear reactors
- qualitative idea of elementary particles

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9.1 INTRODUCTION

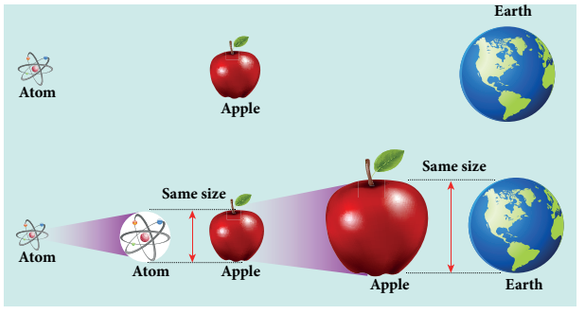


Figure 9.1 Comparision of size of an atom with that of an apple and comparision of size of an apple with that of the Earth

In earlier classes, we have studied that anything which occupies space is called matter. Matter can be classified into solids, liquids and gases. In our daily life, we use water for drinking, petrol for vehicles, we inhale oxygen, stainless steel vessels for cooking, etc. Experiences tell us that behaviour of one material is not the same as that of another, which means that the physical and chemical properties are different for different materials. In order to understand this, we need to know the fundamental constituents of materials.

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